

DATA CLEANING & REPORTING AUTOMATION USING EXCEL, POWER BI AND PYTHON

INTRODUCTION

Data cleaning and reporting automation are important processes in data analytics and business intelligence. Raw datasets often contain duplicate records, missing values, null values, and inconsistent formatting, which can affect business analysis and decision-making. This project focuses on cleaning sales data using Excel, Power BI, and Python. After preprocessing the dataset, interactive dashboards were created in Power BI to analyze sales performance and automate reporting workflows.

OBJECTIVES

- To remove duplicate records from the dataset
- To handle missing values and null values
- To standardize inconsistent data formatting
- To automate data cleaning using Python
- To create interactive dashboards using Power BI
- To improve reporting efficiency and data accuracy

TOOLS USED

Tool	Purpose
Excel	Dataset viewing and inspection
Power BI	Dashboard creation and reporting
Python	Automation of data cleaning
Pandas	Data preprocessing
Openpyxl	Excel file handling

Excel was used to inspect the dataset and understand data quality issues. Power BI was used for data transformation and dashboard development. Python libraries such as Pandas and Openpyxl were used to automate data cleaning and generate cleaned output files.

DATASET DESCRIPTION

Dataset Name: sales_data.xlsx

The dataset contains sales transaction information with columns such as OrderID, Date, Customer, City, Product, Category, Quantity, Price, and PaymentMethod. The dataset contained duplicate rows, missing city values, missing quantity values, and null payment methods. These issues were cleaned before analysis and reporting.

DATA CLEANING PROCESS

Duplicate rows were removed using Power Query in Power BI and the drop_duplicates() function in Python. Missing City and PaymentMethod values were replaced with "Unknown". Missing Quantity values were

replaced with 1 to maintain proper sales calculations. Text formatting inconsistencies were standardized using capitalization techniques. A new column named TotalSales was created using the formula:

TotalSales = Quantity × Price

This column was used for revenue analysis in the dashboard.

DASHBOARD DEVELOPMENT

Interactive dashboards were created in Power BI using various visuals. A Card visual was used to display total sales revenue. A Bar Chart was created to compare city-wise sales performance. A Column Chart was used for product-wise revenue analysis. A Pie Chart was developed to analyze customer payment methods. A Line Chart was used to visualize sales trends over time. Slicers were added for City and Product fields to make the dashboard interactive and user-friendly.

PYTHON AUTOMATION

Python automation was implemented using the Pandas library. The Python script automatically reads the Excel dataset, removes duplicate records, handles missing values, standardizes formatting, creates the TotalSales column, and generates a cleaned Excel output file. This automation reduces manual effort and improves reporting efficiency.

RESULTS

The project successfully improved data quality by removing duplicate records and handling missing values. The Power BI dashboard provided clear visualization of sales performance, product trends, payment behavior, and city-wise revenue. Automation using Python reduced manual preprocessing effort and improved reporting efficiency.

CONCLUSION

This project demonstrated the importance of data cleaning and reporting automation in business analytics. Using Excel, Power BI, and Python, raw sales data was transformed into a clean and structured dataset suitable for analysis. The project improved data consistency, reduced manual reporting work, and enabled interactive visualization of business performance.